Supporting Information for Algorithmic Construction and Exploitation of Topology-Constrained Refinement Models for Semiexperimental Equilibrium Structures

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S1 Global Benchmark Diagnostics

Tables S1 and S2 collect the numerical values underlying the graphical benchmark summaries in the main text.

S2 Atom Numbering Used in the Tables

Figure S1 gives the atom numbering used with the structural tables and diagnostic comparisons. The same Cartesian numbering is used by the topology perception, coordinate generation, and least-squares reporting steps. The p-EBN structure is included as a legacy MSR-import companion case for numbering consistency. For camphor, the atom numbering is defined by the archived Cartesian input and by the full parameter tables in Section S3.

The norcamphor rows use the standalone `xyzin` file archived in the MORPHEUS norcamphor example directory. That file contains the DPCS3 Cartesian geometry, the experimental rotational constants from Table 3 of the norcamphor rotational study, and the tabulated vibrational corrections. The reported run adds 58 Kraitchman-derived soft predicates with $\sigma_R = 0.003$ Å and $\sigma_A = 0.3°$; the same predicate set gives the same residual with GIC and symmetry-adapted Cartesian working coordinates.

The camphor rows report the accepted no-Kraitchman model described in Section S3. The same BDPCS3-predicate model was replayed with symmetry-adapted Cartesian working coordinates as a final-validation check, not as a separate manually constructed refinement.

S3 Camphor Model-Validation Stress Test

The camphor data set is included to document the final validation layer rather than to add another ordinary benchmark. The accepted model uses the BDPCS3 Cartesian geometry as the reference structure, 12 isotopologue records, the three principal moments for each isotopologue, and 178 BDPCS3 soft predicates. No hydrogen coordinate is fixed.

Table S1: One-page audit of the representative `MORPHEUS` semiexperimental refinements. "Model" is the working-coordinate model used for the main row. "Fit" gives the independent rotational components entering the normal equations; for moment-based fits, AB denotes (I_a, I_b). "GIC/A" gives the algorithmically generated non-redundant coordinate count and the totally symmetric active count before constraint projection. "Eff./rank" gives the independent variables after primitive-constraint projection and the final numerical rank. Rotational residuals are RMS/maximum absolute values in MHz over all reported diagnostic constants.

System	PG	Model	Fit	Iso.	GIC/A	Eff./rank	Constraints	Steps	RMS/max
Glycolaldehyde	C_s	GIC	ABC	10	18/12	12/12	none	3/0; min	0.5640/1.7961
Cyclopentadiene	C_{2v}	Cart.	ABC	14	27/10	10/10	none	10/0; min	0.0635/0.2289
Nitrobenzene	C_{2v}	GIC	AB	9	36/13	13/13	8 primitive	4/0; pos.	0.1480/0.2605
p-EBN	C_{2v}	GIC	AB	11	39/14	6/6	22 frozen	4/20; min	0.1190/0.5674
Azulene	C_{2v}	GIC	AB	12	48/17	10/10	8 primitive	8/28; min	0.0731/0.2900
Norcamphor	C_1	GIC	ABC	9	48/48	18/18	30 H + 58 pred.	12/7; min	0.0941/0.2629
Glycine I	C_s	GIC	ABC	5	24/15	11/11	4 function	3/17; min	0.0269/0.0570
Glycine II	C_1	GIC	ABC	10	24/24	19/19	5 function	26/8; min	0.2491/0.6707

Table S2: Planar-component diagnostics for the planar benchmarks. Only two principal moments are independent for a planar molecule; the selected pair marked by an asterisk is used in the normal equations, while all three rotational constants are recomputed after convergence. The residuals are RMS/maximum absolute MHz values over the complete diagnostic A, B, and C constant set.

System	Pair	Fitted moments	Rank	Cond.	Diagnostic RMS/max
Nitrobenzene	AB*	(I_a, I_b)	13	27464.4	0.1480/0.2605
Nitrobenzene	AC	(I_a, I_c)	13	56007.7	0.2593/0.4597
Nitrobenzene	BC	(I_b, I_c)	13	10623.3	2.4602/4.3693
Azulene	AB*	(I_a, I_b)	10	956.7	0.0731/0.2900
Azulene	AC	(I_a, I_c)	10	1780.0	0.1086/0.4222
Azulene	BC	(I_b, I_c)	10	950.2	0.3232/0.7541

Table S3: Comparison of MORPHEUS working-coordinate models. "Work." is the number of non-redundant GICs or symmetry-adapted Cartesian vibrational directions, "Sym." is the number of totally symmetric working coordinates before primitive-constraint projection, and "Eff." is the number of independent least-squares variables after projection. Residuals are RMS/maximum absolute values in MHz.

System	Model	Work.	Sym.	Eff.	Rank	Cond.	Residuals
Cyclopentadiene	GICs	27	10	10	10	454.5	0.0635/0.2289
Cyclopentadiene	Sym. Cart.	27	10	10	10	635.4	0.0635/0.2289
p-EBN	GICs	39	14	6	6	2440.2	0.1190/0.5674
Azulene	GICs	48	17	10	10	956.7	0.0731/0.2900
Azulene	Sym. Cart.	48	17	10	10	744.8	0.0731/0.2900
Norcamphor	GICs	48	48	18	18	5199.2	0.0941/0.2629
Norcamphor	Sym. Cart.	48	48	18	18	4216.7	0.0941/0.2629
Camphor	GICs	75	75	75	75	71.3	0.0788/0.2839
Camphor	Sym. Cart.	75	75	75	75	23.5	0.0788/0.2839

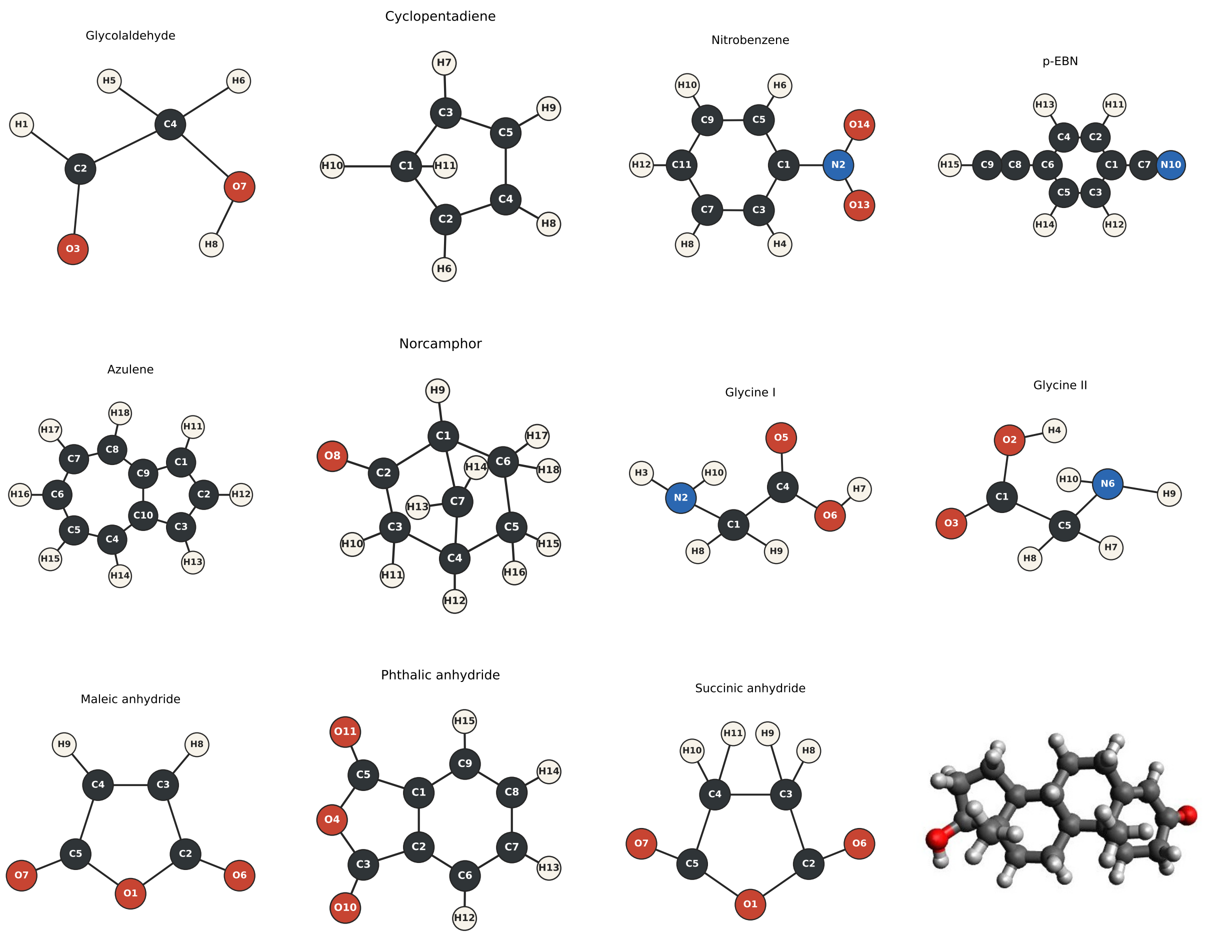

Figure S1: Atom numbering used for the benchmark and companion structures: glycolaldehyde, cyclopentadiene, nitrobenzene, p-EBN, azulene, norcamphor, glycine I, glycine II, and the maleic, phthalic, and succinic anhydride series. Testosterone is included as the 49-atom fused-ring scale check.

C–H distances and hydrogen angles enter as strict soft predicates from the BDPCS3 reference, whereas Kraitchman-derived relations are retained only as diagnostics unless explicitly tested as additional predicates. The files needed to regenerate the tables are archived in `data/camphor`.

Table S4: Camphor accepted no-Kraitchman model. Signed shifts are final SE minus BDPCS3 initial values; propagated errors are standard errors from the final covariance.

Parameter set	Count	Mean shift	RMS shift	Max |shift|	Mean/max error
Bonds / mÅ	28	-0.011	0.314	0.832	2.819/3.000
Angles / deg	57	-0.000	0.015	0.040	0.183/0.257
Dihedrals / deg	93	0.001	0.014	0.041	0.174/0.226

Table S5: Camphor stability controls. Geometry columns report maximum absolute final-minus-BDPCS3 deviations in the corresponding run.

Model variant	RMS / MHz	Rank	Cond.	Max bond / mÅ	Max angle / deg	Outcome
No-Kraitchman BDPCS3 predicates	0.0788	75	71.3	0.832	0.040	retained
Symmetry-Cartesian replay	0.0788	75	23.5	0.832	0.040	same model
All predicate errors x3	0.0727	75	112.5	0.834	0.045	stable
BDPCS3 + broad Kraitchman	0.0780	75	71.1	0.930	0.051	not retained
BDPCS3 + tight Kraitchman	0.0840	75	69.8	7.645	0.343	rejected
No H predicates	0.0629	36	∞	159.509	5.284	rejected
Partial Kraitchman model	0.2415	75	4.04e+04	90.580	6.224	rejected

Table S6: Kraitchman-derived trial predicates for camphor compared with BDPCS3 reference parameters. Full CSV: `data/camphor/kraitchman_predicate_vs_bdpcs3.csv`.

Set	Count	RMS diff.	Max |diff.|	Largest example
Distances / mÅ	28	38.352	92.760	R(13,25) (-92.760)
Angles / deg	57	2.740	6.188	A(25,13,26) (6.188)

The final geometry table shows that the accepted model is a small correction to the BDPCS3 structure: the largest bond and valence-angle shifts are 0.832 mÅ and 0.040°, respectively. The validation controls show why the lower residual obtained when hydrogen predicates are removed is not a better model: the rank is lost and the geometry moves by chemically unacceptable amounts. Similarly, tight Kraitchman-derived predicates distort the structure, while broad Kraitchman predicates are harmless but add no useful information beyond the validated BDPCS3-predicate model. The compact final-validation summary is reported in the main text; the full leave-predicate-group-out, sigma-scan, robust-loss, and multistart run table is archived as `data/camphor/camphor_final_validation_runs.csv`.

Table S7: Legacy MSR/MSROT compatibility checks. For systems where manually constructed legacy inputs are available with the same active information, MORPHEUS reproduces the MSR/MSROT numerical model while generating the coordinate basis and projection operators automatically. Residuals are RMS/maximum absolute rotational residuals in MHz.

System	Legacy comparison	Active information	Residuals
Cyclopentadiene	MSR/MSROT input	same active GIC space	0.0635/0.2289
Azulene	MSR-comparable subset	fixed C–H data; AB pair	0.0731/0.2900

These checks are not additional fitted models: they verify that, at fixed active coordinates and fixed experimental input, the algorithmically generated MORPHEUS model reaches the same numerical result as the corresponding manually prepared MSR/MSROT setup.

S4 Glycine I C_s Constraint Benchmark

Table S9 reports the Gly-I benchmark run from the QC reference Cartesian geometry with no stored structural constraints. The fit uses moments of inertia as observables and all three rotational components because the C_s structure is not a planar rotor in the inertial sense: the two methylene and two amino hydrogens lie outside the heavy-atom plane. The ground-state constants and statistical weights are from Godfrey and Brown, J. Am. Chem. Soc. 117, 2019-2023 (1995), Table 1. The corrected constants are $B_e = B_0 + \Delta B_{\mathrm{vib}}$ with the Gly-Ip zero-point corrections of Kasalová et al. The MORPHEUS advisor generated the following four hard constraints from the QC reference geometry and isotope-availability pattern:

```
HNH(Frozen,Value=105.634453352622)=A(3,2,10)
NH2WAG(Frozen,Value=55.901476176288)=U(1,2,3,10)
ROH(Frozen,Value=0.965159162788)=R(6,7)
ACOH(Frozen,Value=106.584216097335)=A(4,6,7)
```

The numerical data entering the fit are collected in Table S8 so that the corrected constants can be reproduced without consulting the original tables.

Table S8: Glycine I rotational input data. B_0, $\sigma(B_0)$, ΔB_{vib}, and $B_e = B_0 + \Delta B_{\mathrm{vib}}$ are in MHz. The ^{13}C(1) and ^{13}C(2) labels of Godfrey and Brown correspond here to C13_carboxyl and C13_methylene, respectively.

Isotopologue	Constant	B_0	$\sigma(B_0)$	ΔB_{vib}	B_e
parent	A	10341.5210	0.0890	78.6700	10420.1910
parent	B	3876.1785	0.0012	31.3000	3907.4785
parent	C	2912.3509	0.0010	22.3300	2934.6809
C13_carboxyl	A	10340.9430	0.0290	77.8000	10418.7430
C13_carboxyl	B	3869.1142	0.0057	30.9500	3900.0642

Isotopologue	Constant	B_0	$\sigma(B_0)$	ΔB_{vib}	B_e
C13_carboxyl	C	2908.3166	0.0026	22.0600	2930.3766
C13_methylene	A	10227.0840	0.0880	76.7300	10303.8140
C13_methylene	B	3859.0510	0.0210	30.9500	3890.0010
C13_methylene	C	2893.5960	0.0200	21.9200	2915.5160
CD2	A	9318.4520	0.0230	69.8400	9388.2920
CD2	B	3799.2353	0.0023	31.2100	3830.4453
CD2	C	2832.3222	0.0015	21.4800	2853.8022
N15	A	10341.4500	0.1200	78.7600	10420.2100
N15	B	3762.4570	0.0530	30.1300	3792.5870
N15	C	2847.6390	0.0290	21.7100	2869.3490

Table S9: Complete rotational-constant comparison for glycine I. Corrected experimental constants, calculated constants, and differences are in MHz.

Isotopologue	Constant	Corrected exp.	Calculated	Difference
parent	A	10420.1910	10420.2481	-0.0570
parent	B	3907.4785	3907.4786	-0.0001
parent	C	2934.6809	2934.6808	+0.0001
C13_carboxyl	A	10418.7430	10418.7403	+0.0027
C13_carboxyl	B	3900.0642	3900.0634	+0.0008
C13_carboxyl	C	2930.3766	2930.3769	-0.0003
C13_methylene	A	10303.8140	10303.7637	+0.0503
C13_methylene	B	3890.0010	3889.9809	+0.0201
C13_methylene	C	2915.5160	2915.5484	-0.0324
CD2	A	9388.2920	9388.2920	+0.0000
CD2	B	3830.4453	3830.4453	-0.0000
CD2	C	2853.8022	2853.8022	-0.0000
N15	A	10420.2100	10420.2444	-0.0344
N15	B	3792.5870	3792.5436	+0.0434
N15	C	2869.3490	2869.3715	-0.0225

Table S10: Selected primitive structural parameters for the glycine I C_s benchmark with MORPHEUS-generated constraints. Bond lengths are in Å; angles and dihedrals are in degrees.

Parameter	Description	Value	Std. error
R(1,2)	C–N	1.44285	0.00076
R(1,4)	C–C	1.51134	0.00085

Parameter	Description	Value	Std. error
R(1,8)	C–H	1.09081	0.00021
R(1,9)	C–H	1.09081	0.00021
R(2,3)	N–H	1.00228	0.00011
R(2,10)	N–H	1.00228	0.00011
R(4,5)	C–O	1.20529	0.00184
R(4,6)	C–O	1.35001	0.00128
R(6,7)	O–H	0.96516	0.00000
A(2,1,4)	N–C–C	115.266	0.062
A(8,1,9)	H–C–H	105.923	0.031
A(3,2,10)	H–N–H	105.634	0.000
A(1,4,5)	C–C–O	125.338	0.095
A(1,4,6)	C–C–O	111.640	0.123
A(5,4,6)	O–C–O	123.022	0.060
A(4,6,7)	C–O–H	106.584	0.000
D(2,1,4,5)	N–C–C–O	180.000	0.000
D(1,4,6,7)	C–C–O–H	0.000	0.000
D(8,1,4,5)	H–C–C–O	-56.781	0.030
D(9,1,4,5)	H–C–C–O	56.781	0.030

S5 Glycine II Functional-Constraint Benchmark

Table S11 reports the Gly-II advisor-constrained benchmark discussed in the main text. The fit uses moments of inertia as observables, but the diagnostic table is reported in MHz. The corrected experimental constants are $B_e = B_0 + \Delta B_{\mathrm{vib}}$ with the Gly-IIn zero-point corrections. The Cartesian input does not store constraints; the five functional constraints are generated by the MORPHEUS advisor from the QC geometry and the isotopologue coverage.

Table S11: Complete rotational-constant comparison for glycine II. Corrected experimental constants, calculated constants, and differences are in MHz.

Isotopologue	Constant	Corrected exp.	Calculated	Difference
parent	A	10195.5900	10196.2607	-0.6707
parent	B	4105.3670	4105.3251	0.0419
parent	C	3039.8450	3039.8159	0.0291
N15	A	10194.1150	10194.6721	-0.5571
N15	B	3994.8380	3994.8046	0.0334
N15	C	2978.7993	2978.7971	0.0022
C13_methylene	A	10060.9700	10061.6298	-0.6598
C13_methylene	B	4088.6720	4088.5073	0.1647
C13_methylene	C	3018.8730	3018.8495	0.0235

Isotopologue	Constant	Corrected exp.	Calculated	Difference
C13_carboxyl	A	10195.6980	10195.0945	0.6035
C13_carboxyl	B	4093.0000	4093.1241	-0.1241
C13_carboxyl	C	3033.1070	3033.1532	-0.0462
O18_carbonyl	A	10064.2400	10063.7580	0.4820
O18_carbonyl	B	3929.9160	3929.9078	0.0082
O18_carbonyl	C	2931.7130	2931.8037	-0.0907
O18_hydroxyl	A	9580.9720	9580.9886	-0.0166
O18_hydroxyl	B	4090.8498	4090.8965	-0.0467
O18_hydroxyl	C	2975.1370	2975.1377	-0.0007
OD_NH2	A	9792.2000	9792.2302	-0.0302
OD_NH2	B	4096.1490	4096.1104	0.0386
OD_NH2	C	2998.1238	2998.0987	0.0251
CD2	A	9145.3030	9145.3053	-0.0023
CD2	B	4020.3863	4020.3836	0.0027
CD2	C	2947.9923	2947.9913	0.0010
OH_ND2	A	9927.6990	9927.6852	0.0138
OH_ND2	B	3716.3758	3716.3613	0.0145
OH_ND2	C	2841.7341	2841.7204	0.0137
OD_ND2	A	9542.6600	9542.6581	0.0019
OD_ND2	B	3710.5050	3710.5504	-0.0454
OD_ND2	C	2806.0910	2806.1569	-0.0659

S6 Cyclopentadiene Benchmark

Table S12 reports the complete comparison between corrected experimental rotational constants and constants calculated from the final semiexperimental structure. The workflow uses generalized internal coordinates (GICs) or symmetry-adapted Cartesian coordinates generated within `MORPHEUS`. The data set contains 14 isotopologues and all three principal rotational constants for each isotopologue. The MORPHEUS refinement uses only a Cartesian representation of the molecular structure, whereas the MSR refinement used manual Z-matrix coordinates and constraints.

Table S12: Complete rotational-constant comparison for the cyclopentadiene benchmark derived from MSR data. Corrected experimental and calculated constants are in MHz.

Isotopologue	Component	Corrected exp.	Calculated	Difference
msr_symm_01	A	8489.154993	8489.232117	-0.077124
msr_symm_01	B	8292.100993	8292.093033	0.007960
msr_symm_01	C	4305.664997	4305.658417	0.006580
msr_symm_02	A	8292.098993	8292.093033	0.005960

Isotopologue	Component	Corrected exp.	Calculated	Difference
msr_symm_02	B	8280.278992	8280.319224	-0.040231
msr_symm_02	C	4251.270992	4251.257436	0.013556
msr_symm_03	A	8482.790995	8482.866229	-0.075234
msr_symm_03	B	8105.052002	8105.082639	-0.030637
msr_symm_03	C	4253.082997	4253.084517	-0.001520
msr_symm_04	A	8408.381002	8408.358013	0.022989
msr_symm_04	B	8172.557990	8172.606506	-0.048516
msr_symm_04	C	4252.643011	4252.628488	0.014523
msr_symm_05	A	8408.381002	8408.358024	0.022978
msr_symm_05	B	8172.557990	8172.606496	-0.048506
msr_symm_05	C	4252.643011	4252.628488	0.014523
msr_symm_06	A	8195.611995	8195.593227	0.018768
msr_symm_06	B	7918.397999	7918.289812	0.108187
msr_symm_06	C	4178.352982	4178.398406	-0.045424
msr_symm_07	A	8195.611995	8195.612584	-0.000589
msr_symm_07	B	7918.397999	7918.308530	0.089469
msr_symm_07	C	4178.352982	4178.398587	-0.045604
msr_symm_08	A	8477.175001	8477.228373	-0.053372
msr_symm_08	B	7650.516996	7650.483876	0.033120
msr_symm_08	C	4123.171991	4123.147057	0.024934
msr_symm_09	A	8477.175001	8477.228367	-0.053366
msr_symm_09	B	7650.516996	7650.483883	0.033113
msr_symm_09	C	4123.171991	4123.147057	0.024933
msr_symm_10	A	8371.829003	8371.953330	-0.124328
msr_symm_10	B	7735.107991	7735.111002	-0.003010
msr_symm_10	C	4122.272002	4122.241235	0.030767
msr_symm_11	A	8371.829003	8371.953344	-0.124341
msr_symm_11	B	7735.107991	7735.110990	-0.002998
msr_symm_11	C	4122.272002	4122.241235	0.030767
msr_symm_12	A	7057.547996	7057.656561	-0.108565
msr_symm_12	B	6730.362994	6730.334074	0.028920
msr_symm_12	C	3555.255987	3555.260397	-0.004410
msr_symm_13	A	7057.547996	7057.672705	-0.124709
msr_symm_13	B	6730.362994	6730.349532	0.013462
msr_symm_13	C	3555.255987	3555.260180	-0.004193
msr_symm_14	A	6656.949995	6656.991310	-0.041315
msr_symm_14	B	6653.764996	6653.536122	0.228874
msr_symm_14	C	3469.286997	3469.303412	-0.016415

S7 Nitrobenzene Planar Benchmark

Nitrobenzene is a planar aromatic benchmark. The least-squares fit used the independent AB moment pair, corresponding to (I_a, I_b), while the C constants were recomputed from the optimized Cartesian geometry and retained as diagnostic quantities. Table S13 reports the complete diagnostic comparison.

Table S13: Complete rotational-constant comparison for the nitrobenzene planar benchmark. Corrected experimental and calculated constants are in MHz. The fit used the independent AB pair; the C component is diagnostic.

Isotopologue	Component	Corrected exp.	Calculated	Difference
parent	A	3995.209000	3995.211821	-0.002821
parent	B	1296.305300	1296.307106	-0.001806
parent	C	979.000500	978.740045	+0.260455
N15	A	3995.193000	3995.211821	-0.018821
N15	B	1287.445000	1287.445036	-0.000036
N15	C	973.938200	973.679678	+0.258522
O18	A	3925.992000	3925.984021	+0.007979
O18	B	1264.689700	1264.689116	+0.000584
O18	C	956.803400	956.552095	+0.251305
6C13	A	3995.172000	3995.211821	-0.039821
6C13	B	1274.812600	1274.812638	-0.000038
6C13	C	966.688800	966.436980	+0.251820
6D	A	3995.241000	3995.211821	+0.029179
6D	B	1253.544800	1253.544803	-0.000003
6D	C	954.416100	954.164457	+0.251643
2C13	A	3949.267000	3949.255685	+0.011315
2C13	B	1295.514600	1295.514106	+0.000494
2C13	C	975.768500	975.508297	+0.260203
2D	A	3856.605000	3856.604007	+0.000993
2D	B	1296.294600	1296.294437	+0.000163
2D	C	970.447500	970.190734	+0.256766
3C13	A	3950.485000	3950.473551	+0.011449
3C13	B	1284.693900	1284.693395	+0.000505
3C13	C	969.688400	969.433703	+0.254697
3D	A	3858.465000	3858.463986	+0.001014
3D	B	1276.881200	1276.881052	+0.000148
3D	C	959.645200	959.390171	+0.255029

The Kraitchman diagnostic for nitrobenzene is reported in Table S14. The comparison uses absolute substitution coordinates because the Kraitchman equations do not determine coordinate signs. The diagnostic was not used in the least-squares objective.

Table S14: Kraitchman diagnostic for the nitrobenzene planar benchmark. Absolute Kraitchman substitution coordinates and absolute fitted coordinates are in Å.

Isotopologue	Atom	Axis	Kraitchman	Fit	Difference
N15	2 N	a	1.64701	1.64723	-0.00022
N15	2 N	b	0.01795	0.00000	+0.01795
N15	2 N	c	0.01379	0.00000	+0.01379
O18	13 O	a	2.22302	2.21369	+0.00933
O18	13 O	b	1.06338	1.08300	-0.01962
O18	13 O	c	0.00896	0.00000	+0.00896
6C13	11 C	a	2.56993	2.56999	-0.00005
6C13	11 C	b	0.03786	0.00000	+0.03786
6C13	11 C	c	0.00000	0.00000	+0.00000
6D	12 H	a	3.65015	3.65027	-0.00012
6D	12 H	b	0.00000	0.00000	+0.00000
6D	12 H	c	0.01902	0.00000	+0.01902
2C13	3 C	a	0.48915	0.48830	+0.00085
2C13	3 C	b	1.21604	1.21671	-0.00066
2C13	3 C	c	0.00000	0.00000	+0.00000
2D	4 H	a	0.05793	0.06125	-0.00332
2D	4 H	b	2.13423	2.13424	-0.00001
2D	4 H	c	0.00000	0.00000	+0.00000
3C13	7 C	a	1.88184	1.87669	+0.00515
3C13	7 C	b	1.19989	1.20775	-0.00786
3C13	7 C	c	0.00000	0.00000	+0.00000
3D	8 H	a	2.43739	2.41687	+0.02052
3D	8 H	b	2.11909	2.14308	-0.02399
3D	8 H	c	0.03090	0.00000	+0.03090

S8 p-EBN Legacy MSR-Import Benchmark

The p-EBN case was recovered from the legacy MSR-import fixtures and rerun with the current MORPHEUS workflow. The current run imports the Z-matrix geometry, isotopologue definitions, rotational constants, vibrational corrections, weights, and frozen Z-matrix parameters. The historical Z-matrix-style equality-constraint block is not part of the current public MORPHEUS constraint API and was therefore not imposed in this compatibility run. Table S15 summarizes the fit, and Tables S16–S18 report the rotational constants, Kraitchman diagnostic, and selected structural parameters.

Table S15: p-EBN fit diagnostics from the legacy MSR-import benchmark. The fit used the independent AB moment pair; the C rotational constants are retained as diagnostics.

Diagnostic	Value
Isotopologues	11
Effective rank	6
Effective parameters	6
Condition number	2440.25
Rotational RMS / MHz	0.118975
Max $\lvert \Delta B \rvert$ / MHz	0.567403
Weighted RMS	0.141739
Stationary point	minimum

Table S16: Complete rotational-constant comparison for the p-EBN legacy MSR-import benchmark. Corrected experimental and calculated constants are in MHz. The fit used the independent AB moment pair; C is diagnostic.

Isotopologue	Component	Corrected exp.	Calculated	Difference
parent	A	5690.084365	5690.075486	0.008879
parent	B	711.461697	711.463764	-0.002068
parent	C	632.404642	632.392049	0.012593
^{13}C(1)	A	5689.787883	5690.075486	-0.287603
^{13}C(1)	B	709.591078	709.571546	0.019532
^{13}C(1)	C	630.926783	630.896616	0.030166
^{13}C(2)	A	5598.029883	5598.023971	0.005913
^{13}C(2)	B	711.004858	711.013323	-0.008464
^{13}C(2)	C	630.892369	630.883832	0.008537
^{13}C(3)	A	5598.029883	5598.023627	0.006256
^{13}C(3)	B	711.004858	711.013348	-0.008490
^{13}C(3)	C	630.892369	630.883848	0.008521
^{13}C(4)	A	5598.343855	5598.493203	-0.149349
^{13}C(4)	B	710.960437	710.963815	-0.003379
^{13}C(4)	C	630.861335	630.850813	0.010522
^{13}C(5)	A	5598.343855	5598.496086	-0.152231
^{13}C(5)	B	710.960437	710.963789	-0.003352
^{13}C(5)	C	630.861335	630.850828	0.010506
^{13}C(6)	A	5690.100926	5690.075486	0.025441
^{13}C(6)	B	709.476356	709.476380	-0.000024
^{13}C(6)	C	630.835668	630.821383	0.014285
^{13}C(7)	A	5689.975800	5690.075486	-0.099685
^{13}C(7)	B	703.698492	703.695775	0.002716

Isotopologue	Component	Corrected exp.	Calculated	Difference
^{13}C(7)	C	626.264052	626.247314	0.016738
^{13}C(8)	A	5690.022772	5690.075486	-0.052713
^{13}C(8)	B	703.494496	703.497679	-0.003183
^{13}C(8)	C	626.102657	626.090418	0.012238
^{13}C(9)	A	5690.053400	5690.075486	-0.022086
^{13}C(9)	B	695.483305	695.491125	-0.007820
^{13}C(9)	C	619.748527	619.740932	0.007596
^{15}N(10)	A	5690.642889	5690.075486	0.567403
^{15}N(10)	B	696.225258	696.222764	0.002494
^{15}N(10)	C	620.336151	620.321809	0.014342

Table S17: Kraitchman diagnostic for the p-EBN legacy MSR-import benchmark. Absolute Kraitchman substitution coordinates and absolute fitted coordinates are in Å.

Isotopologue	Atom	Axis	Kraitchman	Fit	Difference
^{13}C(1)	1 C	*a*	1.37054	1.37943	-0.00889
^{13}C(1)	1 C	*b*	0.04432	0.00001	0.04431
^{13}C(1)	1 C	*c*	0.05182	0.00000	0.05182
^{13}C(2)	2 C	*a*	0.67660	0.67156	0.00505
^{13}C(2)	2 C	*b*	1.21097	1.21167	-0.00070
^{13}C(2)	2 C	*c*	0.02620	0.00000	0.02620
^{13}C(3)	3 C	*a*	0.67660	0.67154	0.00506
^{13}C(3)	3 C	*b*	1.21097	1.21168	-0.00071
^{13}C(3)	3 C	*c*	0.02620	0.00000	0.02620
^{13}C(4)	4 C	*a*	0.70882	0.70753	0.00130
^{13}C(4)	4 C	*b*	1.20888	1.20858	0.00030
^{13}C(4)	4 C	*c*	0.02564	0.00000	0.02564
^{13}C(5)	5 C	*a*	0.70882	0.70755	0.00128
^{13}C(5)	5 C	*b*	1.20888	1.20856	0.00032
^{13}C(5)	5 C	*c*	0.02564	0.00000	0.02564
^{13}C(6)	6 C	*a*	1.41308	1.41379	-0.00071
^{13}C(6)	6 C	*b*	0.00000	0.00000	0.00000
^{13}C(6)	6 C	*c*	0.00000	0.00000	0.00000
^{13}C(7)	7 C	*a*	2.80547	2.80656	-0.00109
^{13}C(7)	7 C	*b*	0.02098	0.00001	0.02098
^{13}C(7)	7 C	*c*	0.03553	0.00000	0.03553
^{13}C(8)	8 C	*a*	2.84256	2.84253	0.00003
^{13}C(8)	8 C	*b*	0.00000	0.00000	0.00000
^{13}C(8)	8 C	*c*	0.03179	0.00000	0.03179
^{13}C(9)	9 C	*a*	4.04886	4.04814	0.00072

Isotopologue	Atom	Axis	Kraitchman	Fit	Difference
^{13}C(9)	9 C	b	0.01219	0.00000	0.01219
^{13}C(9)	9 C	c	0.01835	0.00000	0.01835
^{15}N(10)	10 N	a	3.96474	3.96467	0.00007
^{15}N(10)	10 N	b	0.00000	0.00000	0.00000
^{15}N(10)	10 N	c	0.00000	0.00000	0.00000

Table S18: Selected structural parameters from the p-EBN MORPHEUS fit. Bond lengths are in Å; angles are in degrees.

Parameter	Description	Value	Std. error
R(1,2)	C–C	1.40330	0.01208
R(1,3)	C–C	1.40330	0.01208
R(1,7)	C–C	1.42713	0.04664
R(2,4)	C–C	1.37909	0.03102
R(2,11)	C–H	1.08012	0.00000
R(3,5)	C–C	1.37909	0.03102
R(3,12)	C–H	1.08012	0.00000
R(4,6)	C–C	1.39981	0.00002
R(4,13)	C–H	1.08006	0.00000
R(5,6)	C–C	1.39979	0.00002
R(5,14)	C–H	1.08006	0.00000
R(6,8)	C–C	1.42873	0.03574
R(7,10)	C–N	1.15811	0.03755
R(8,9)	C–C	1.20562	0.03576
R(9,15)	C–H	1.06154	0.00000
A(2,1,3)	C–C–C	119.411	1.873
A(2,1,7)	C–C–C	120.294	0.936
A(3,1,7)	C–C–C	120.295	0.936
A(1,2,4)	C–C–C	120.165	0.983
A(1,2,11)	C–C–H	119.674	0.000
A(4,2,11)	C–C–H	120.161	0.983
A(1,3,5)	C–C–C	120.165	0.983
A(1,3,12)	C–C–H	119.674	0.000
A(5,3,12)	C–C–H	120.161	0.983
A(2,4,6)	C–C–C	120.429	0.046
A(2,4,13)	C–C–H	120.212	0.046
A(6,4,13)	C–C–H	119.359	0.000
A(3,5,6)	C–C–C	120.430	0.046
A(3,5,14)	C–C–H	120.211	0.046
A(6,5,14)	C–C–H	119.359	0.000
A(4,6,5)	C–C–C	119.398	0.001

Parameter	Description	Value	Std. error
A(4,6,8)	C–C–C	120.301	0.000
A(5,6,8)	C–C–C	120.301	0.000
A(1,7,10)	C–C–N	180.000	0.000
A(6,8,9)	C–C–C	180.000	0.000
A(8,9,15)	C–C–H	180.000	0.000

S9 Azulene Comparison with the Reference Refinement

The azulene calculation used fixed C1–H11, C3–H13, and C2–H12 bond lengths; the remaining C–H distances are refined. All internal coordinates were generated algorithmically from Cartesian structures. Because azulene is planar, the fit used the independent AB moment pair and retained the C constants as diagnostics.

Table S19: Complete rotational-constant comparison for the azulene MSR-comparable subset. Corrected experimental and calculated constants are in MHz.

Isotopologue	Component	Corrected exp.	Calculated	Difference
parent	A	2862.097939	2862.075372	+0.022567
parent	B	1262.517835	1262.559239	-0.041404
parent	C	876.138923	876.087228	+0.051695
azulene-4-d1	A	2752.561850	2752.579320	-0.017470
azulene-4-d1	B	1262.294384	1262.218223	+0.076161
azulene-4-d1	C	865.456501	865.387542	+0.068959
azulene-5-d1	A	2794.488840	2794.483227	+0.005613
azulene-5-d1	B	1241.273362	1241.275392	-0.002030
azulene-5-d1	C	859.559161	859.497208	+0.061953
azulene-6-d1	A	2862.003650	2862.075372	-0.071722
azulene-6-d1	B	1223.400793	1223.394588	+0.006205
azulene-6-d1	C	857.105233	857.048897	+0.056336
azulene-4,8-d2	A	2649.532362	2649.523242	+0.009120
azulene-4,8-d2	B	1262.005103	1261.903897	+0.101206
azulene-4,8-d2	C	854.871156	854.788696	+0.082460
azulene-5,7-d2	A	2725.125973	2725.133539	-0.007566
azulene-5,7-d2	B	1221.780703	1221.778074	+0.002629
azulene-5,7-d2	C	843.602156	843.573085	+0.029071
azulene-1-13C	A	2841.331150	2841.459456	-0.128306
azulene-1-13C	B	1251.448996	1251.449765	-0.000769
azulene-1-13C	C	868.849088	868.805922	+0.043166
azulene-2-13C	A	2862.093080	2862.075372	+0.017708

Isotopologue	Component	Corrected exp.	Calculated	Difference
azulene-2-13C	B	1240.233194	1240.296442	-0.063248
azulene-2-13C	C	865.322674	865.309646	+0.013028
azulene-9-13C	A	2853.194180	2852.904191	+0.289989
azulene-9-13C	B	1261.606550	1261.641703	-0.035153
azulene-9-13C	C	874.835186	874.784969	+0.050217
azulene-4-13C	A	2821.753737	2821.770671	-0.016934
azulene-4-13C	B	1261.608919	1261.591491	+0.017428
azulene-4-13C	C	871.875087	871.811446	+0.063641
azulene-5-13C	A	2836.950530	2837.099313	-0.148783
azulene-5-13C	B	1251.104248	1251.070937	+0.033311
azulene-5-13C	C	868.277606	868.215431	+0.062175
azulene-6-13C	A	2862.033150	2862.075372	-0.042222
azulene-6-13C	B	1243.247466	1243.219286	+0.028180
azulene-6-13C	C	866.765213	866.731282	+0.033931

Table S20: Complete structural comparison between MSR and MORPHEUS results for azulene employing analogous constraints. Bond lengths are in Å; angles are in degrees.

Parameter	Unit	`MORPHEUS`	σ	MSR	Difference
$R(1,9)$	Å	1.39745	0.00599	1.39900	-0.00155
$R(4,10)$	Å	1.37923	0.01201	1.37800	+0.00123
$R(4,5)$	Å	1.40122	0.00653	1.40100	+0.00022
$R(5,6)$	Å	1.39054	0.00442	1.39200	-0.00146
$R(4,14)$	Å	1.08356	0.00220	1.08700	-0.00344
$R(5,15)$	Å	1.08133	0.00230	1.08200	-0.00067
$R(6,16)$	Å	1.08117	0.00441	1.08200	-0.00083
$R(1,11)$	Å	1.08116	0.00000	1.08000	+0.00116
$R(2,12)$	Å	1.08236	0.00000	1.08100	+0.00136
$R(1,2)$	Å	1.39593	0.00584	1.40400	-0.00807
$R(9,10)$	Å	1.50860	0.01549	1.50000	+0.00860
$A(1,9,10)$	degree	106.13001	0.52104	106.50000	-0.36999
$A(5,6,7)$	degree	130.09492	0.17695	129.90000	+0.19492
$A(6,5,15)$	degree	115.90333	0.30686	115.70000	+0.20333
$A(4,5,6)$	degree	128.68339	0.00000	128.68000	+0.00339
$A(5,4,10)$	degree	128.79958	0.00000	128.81000	-0.01042
$A(9,1,11)$	degree	125.13843	0.00000	125.15000	-0.01157
$A(10,4,14)$	degree	115.31629	0.00000	115.31000	+0.00629
$A(1,2,12)$	degree	125.06740	0.00000	125.07000	-0.00260
$A(1,2,3)$	degree	109.86520	0.00000	109.71000	+0.15520

Table S21: Summary of the anhydride literature-data tests. "GICs" is the number of final non-redundant coordinates generated by MORPHEUS, "Act." is the number of active totally symmetric coordinates before projection of hard constraints, and "Eff." is the final rank/effective number of fitted variables. Residuals are RMS/maximum absolute rotational residuals in MHz.

System	Auxiliary relation	GICs	Act.	Eff.	Cond.	Residuals
Maleic anhydride	none	21	8	8	839	0.0359/0.1453
Phthalic anhydride	weighted r_e^{BO} predicates	39	14	14	4.05	0.5013/0.8482
Succinic anhydride	fixed local H frame	27	9	5	498	0.0353/0.0731

Parameter	Unit	`MORPHEUS`	σ	MSR	Difference
$A(2,1,9)$	degree	108.93739	0.52104	108.66000	+0.27739
$A(4,10,9)$	degree	127.46958	0.08847	127.58000	-0.11042
$A(5,6,16)$	degree	114.95254	0.08847	115.07000	-0.11746

S10 Anhydride Literature-Data Series

The three anhydrides form a homologous series used to test how `MORPHEUS` handles progressively less complete experimental information. In all cases the structural refinement starts from a Cartesian parent geometry and tabulated rotational constants. When B_0 and ΔB_{vib} are available, the corrected semiexperimental observables are defined directly as $B_e = B_0 + \Delta B_{\mathrm{vib}}$, with any electronic correction added as a separate term when reported. No line list, SPFIT input, or molecule-specific coordinate model is required by the refinement step.

Maleic anhydride is the complete-data limiting case: the isotopic set is rich enough that no auxiliary structural relation is needed. Phthalic anhydride uses weighted primitive-coordinate predicates because the literature analysis itself treats the r_e^{BO} distances and angles as mixed-estimation information. Succinic anhydride has only heavy-atom substitutions; the hydrogen frame is therefore unsupported by direct isotope information. The production comparison in Table S21 uses fixed local hydrogen parameters. Runs with only soft C–H predicates reproduce the intended local predicate values but can enter a nonphysical hydrogen-frame valley before final topology reporting, which is the expected distinction between soft observations and hard removal of unsupported directions.

S10.1 Maleic Anhydride Literature-Data Reproduction

Maleic anhydride was added as a reproducibility test based entirely on the published semiexperimental dataset of Vogt, Demaison, and Rudolph. The starting geometry was reconstructed from their reported r_e^{SE} parameters, and the ten isotopologue records use the published B_0 constants and rovibrational corrections. The small electronic correction was included by using the tabulated B_e values, equivalently as $\Delta B_{\mathrm{el}} = B_e - B_0 - \Delta B_{\mathrm{vib}}$. No Z-matrix, manual coordinate model, constraints, predicates, or parameter classes were used.

Table S22: Algorithmically generated symmetry-coordinate model for maleic anhydride from literature data. Only the totally symmetric A_1 block is active in the C_{2v} semiexperimental refinement.

Symmetry block	Coordinates	Role in the SE model
A_1	8	Active stretches/bends; rank 8
A_2	3	Torsion/out-of-plane diagnostics
B_1	3	Torsion/out-of-plane diagnostics
B_2	7	Stretch/bend diagnostics
Total	21	Non-redundant GIC space

S10.2 Input and Manifest Excerpt

The complete example distributed with the code consists only of the parent Cartesian geometry and the isotopologue records. The command used for the calculation was

```
python -m matrix semiexp \
  --xyz examples/semiexp/maleic_anhydride/parent.xyz \
  --observations examples/semiexp/maleic_anhydride/isotopologues.toml \
  --outdir working/semiexp/maleic_anhydride \
  --coordinate-model gic \
  --observable moments \
  --rotational-components AB \
  --robust-loss huber
```

The starting geometry was the following Cartesian file, reconstructed from the published semiexperimental structure and used only as an initial point:

```
9
maleic anhydride rse start reconstructed from Vogt Demaison Rudolph Struct Chem
    2011 Table 1
O 0.0000000000 0.0000000000 0.0000000000
C 1.1240088423 -0.8088727727 0.0000000000
C 0.6662000000 -2.2214374877 0.0000000000
C -0.6662000000 -2.2214374877 0.0000000000
C -1.1240088423 -0.8088727727 0.0000000000
O 2.2290423505 -0.3689031082 0.0000000000
O -2.2290423505 -0.3689031082 0.0000000000
H 1.3567205284 -3.0472907736 0.0000000000
H -1.3567205284 -3.0472907736 0.0000000000
```

Each isotopologue record contains the observed B_0 constants and the corrections needed to form the effective equilibrium constants. For example, the parent species is represented as

```
[[isotopologues]]
label = "Main"
substitutions = ""
[isotopologues.constants]
A_MHz = 6842.489000
```

```
B_MHz = 2467.614000
C_MHz = 1813.628000
[isotopologues.vibrational_correction]
delta_A_MHz = 49.780000
delta_B_MHz = 14.180000
delta_C_MHz = 11.160000
convention = "add"
[isotopologues.electronic_correction]
delta_A_MHz = 0.421000
delta_B_MHz = 0.066000
delta_C_MHz = 0.012000
convention = "add"
```

and substituted species are specified by isotope substitutions rather than by new structural models, for example

```
[[isotopologues]]
label = "13C2d2"
substitutions = "2:13;8:2;9:2"
```

The full output manifest is archived with the example. The excerpt below shows the information needed to audit model construction and the absence of manual constraints:

```
[method]
program = MORPHEUS
method = semiexperimental equilibrium-geometry least squares
coordinate_model = gic
coordinate_basis = MORPHEUS non-redundant symmetry-adapted GICs; active subspace
   is totally symmetric
observable = moments
components = Ia,Ib
isotopologues = Main, d2, 13C2, 13C3, 18O1, 18O6, 13C2d2, 13C3d2, 18O1d2, 18O6d2

[constraints]
fixed_parameter = none
parameter_class = none
qm_predicate = none

[constraint_diagnostics]
input_records = 0
input_expression_constraints = 0
input_coordinate_definitions = 0
symmetry_expanded_fixed_primitives = 0
parameter_classes = 0
```

The final diagnostic excerpt reports the rank-consistent fitted model:

```
[fit_statistics]
convergence = step_tolerance
iterations = 71
```

Table S23: Maleic anhydride refinement diagnostics. The GIC fit used moments of inertia for the independent AB planar pair; the omitted C constants are reported as diagnostics.

Quantity	Value
Generated non-redundant GICs	21
Totally symmetric A_1 coordinates	8
Independent fitted variables	8
Rank	8
Condition number	838.856
Rotational RMS residual / MHz	0.03593
Maximum absolute residual / MHz	0.14526
Iterations	71

```
n_optimized_parameters = 8
rank = 8
condition_number = 838.856024878
robust_loss = huber
coordinate_builder_calls = 1

[rotational_residual_statistics]
n_rotational_constants = 30
rotational_rms_MHz = 0.0359332113823
rotational_max_abs_MHz = 0.145260439303

[working_coordinates]
coordinate_count = 21
index active class value sigma label
1 yes - 3.22390949563 0.000251446997267 GIC001 MORPHEUS A1Str0001 irrep=A1
...
8 yes - -0.00433581007571 0.000311593700452 GIC008 MORPHEUS A1Ang0003 irrep=A1
9 no - -2.02080870849 0 GIC009 MORPHEUS A2Tor0001 irrep=A2
...
21 no - 6.53687463342e-16 0 GIC021 MORPHEUS B2Ang0003 irrep=B2
```

Table S24: Complete rotational-constant comparison for maleic anhydride. Corrected experimental and calculated constants are in MHz. The corrected experimental constants reproduce the published B_e values.

Isotopologue	Constant	Corrected exp.	Calculated	Difference
Main	A	6892.6900	6892.7047	-0.0147
Main	B	2481.8600	2481.8511	+0.0089
Main	C	1824.8000	1824.7976	+0.0024
d_2	A	6148.0000	6148.0220	-0.0220

Isotopologue	Constant	Corrected exp.	Calculated	Difference
d_2	B	2437.5100	2437.5096	+0.0004
d_2	C	1745.4800	1745.4787	+0.0013
^{13}C2	A	6891.4300	6891.4558	-0.0258
^{13}C2	B	2466.6500	2466.6492	+0.0008
^{13}C2	C	1816.4800	1816.4793	+0.0007
^{13}C3	A	6739.8700	6739.8341	+0.0359
^{13}C3	B	2476.4200	2476.4210	-0.0010
^{13}C3	C	1811.0000	1811.0031	-0.0031
^{18}O1	A	6738.2200	6738.1958	+0.0242
^{18}O1	B	2481.8500	2481.8511	-0.0011
^{18}O1	C	1813.7800	1813.7867	-0.0067
^{18}O6	A	6839.8800	6839.9516	-0.0716
^{18}O6	B	2367.8700	2367.8784	-0.0084
^{18}O6	C	1758.9400	1758.9566	-0.0166
^{13}C2d_2	A	6146.3100	6146.1647	+0.1453
^{13}C2d_2	B	2422.8100	2422.8401	-0.0301
^{13}C2d_2	C	1737.7900	1737.7951	-0.0051
^{13}C3d_2	A	6033.8400	6033.8538	-0.0138
^{13}C3d_2	B	2432.2700	2432.2723	-0.0023
^{13}C3d_2	C	1733.4900	1733.4936	-0.0036
^{18}O1d_2	A	6013.3300	6013.3614	-0.0314
^{18}O1d_2	B	2437.5200	2437.5096	+0.0104
^{18}O1d_2	C	1734.4500	1734.4516	-0.0016
^{18}O6d_2	A	6099.5800	6099.4994	+0.0806
^{18}O6d_2	B	2327.3500	2327.3468	+0.0032
^{18}O6d_2	C	1684.5800	1684.5745	+0.0055

S11 Testosterone Large-Molecule Scale Check

The testosterone calculation in the main text is not an isotope-rich semiexperimental structure determination. It is a scale and automation check using the published data of Uribe et al. for the detected T1-H2^{-}-gp conformer. Its purpose is to document the full large-molecule chain: the QC protocol provides an accurate equilibrium reference geometry, the accelerated ΔB_{vib} workflow makes vibrational corrections available at this size, and MORPHEUS adds the algorithmic refinement-model layer. The vibrational corrections used here were generated with the finite-difference-of-gradients protocol of Mendolicchio et al. (PCCP 2025); the algorithmic details of that protocol are not rederived in the present work. The only experimental input is the parent ground-state rotational constant set. The semiexperimental target is obtained by subtracting the B3 vibrational corrections reported in the same table:

$$B_e^{\mathrm{SE}} = B_0 - \Delta B_{\mathrm{vib}}.$$

Table S25: Selected maleic anhydride structural parameters from the MORPHEUS reproduction compared with the published r_e^{SE} structure. Bond lengths are in Å; angles are in degrees.

Parameter	Atoms	MORPHEUS	Published	Difference
O–C	1–2	1.38489	1.3848	+0.00009
C–C	2–3	1.48479	1.4849	-0.00011
C=C	3–4	1.33241	1.3324	+0.00001
C=O	2–6	1.18945	1.1894	+0.00005
C–H	3–8	1.07652	1.0765	+0.00002
C–O–C	2–1–5	108.513	108.52	-0.007
O–C–C	1–2–3	107.784	107.78	+0.004
O–C–O	1–2–6	122.537	122.55	-0.013
C–C–O	3–2–6	129.679	129.67	+0.009
C–C–H	4–3–8	129.896	129.90	-0.004

Thus the data used by MORPHEUS are $B_0 = (801.5, 171.9, 156.6)$ MHz and $\Delta B_{\mathrm{vib}} = (8.1, 1.6, 1.4)$ MHz for (A, B, C), giving $B_e^{\mathrm{SE}} = (793.4, 170.3, 155.2)$ MHz.

The calculation starts from the QC Cartesian geometry and does not require a Z-matrix, hand-selected internal coordinates, or a stored refinement model. Topology perception identifies the $C_{19}H_{28}O_2$ molecular graph, MORPHEUS builds the 141-coordinate non-redundant GIC space, and the MORPHEUS class layer applies the reduced-dimensionality model after symmetry and rank reduction. For the baseline parent-only diagnostic, all angular, torsional, and out-of-plane GIC families are fixed. Stretch combinations involving C–H, O–H, and C–O primitives are also fixed. The retained C–C skeletal stretch block is then assigned to one shared parameter class, so the fit contains one effective variable. This is a hard class constraint: it does not add an observation or a statistical prior, but replaces the selected active coordinates by a single shared displacement direction before the least-squares normal equations are formed.

Two further models were generated with the same automated basis. In the two-class model, the refinement model is specified through the C–C skeletal class and the C–O stretch class as primitive bond sets. MORPHEUS then maps the generated GICs onto disjoint shared classes with default coefficient thresholds. Later, more specific classes in the declaration have priority over earlier, broader classes; GICs that do not satisfy the thresholds, or that cannot be supported within the data-compatible class budget, are frozen. In the predicate model, a compact C–C/C–O GIC subset is kept independent and XY QC bond lengths for the corresponding graph are added as weighted observations with a default uncertainty of 0.005 Å. A second sensitivity run used 0.01 Å. These values are tied to the expected quality of the QM reference geometry: 0.005–0.01 Å is already a conservative range for good equilibrium XY bonds, whereas 0.05 Å would be too loose to represent a chemically meaningful bond-length predicate. The predicate uncertainty is exposed as `MATRIX_TESTOSTERONE_XY_PREDICATE_SIGMA`, because it controls the tradeoff between the parent rotational data and the QC reference geometry.

Table S26: Testosterone T1-H2$^-$-gp rotational constants used for the large-molecule scale check. Constants are in MHz. The QC column is the unrefined rotational constant calculated from the input geometry. The MORPHEUS columns are, respectively, the one-class C–C baseline, the two-class C–C/C–O model, and weighted XY-predicate sensitivity runs.

Constant	B_0	ΔB_{vib}	B_e^{SE}	QC	C–C	C–C/C–O	Pred. 0.005	Pred. 0.01
A	801.5	8.1	793.4	788.762	793.427	793.398	789.272	790.286
B	171.9	1.6	170.3	169.626	169.667	170.254	169.596	169.545
C	156.6	1.4	155.2	154.637	154.908	155.258	154.645	154.669

Table S27: Testosterone model-construction diagnostics. All calculations used rotational constants as observables and all three parent components. The rotational RMS residual is in MHz.

Quantity	C–C	C–C/C–O	Pred. 0.005	Pred. 0.01
Atoms	49	49	49	49
Non-redundant GICs	141	141	141	141
Retained C–C GICs	5	21	5	5
Retained C–O GICs	0	2	2	2
Effective fitted variables	1	2	7	7
Rank	1	2	7	7
Condition number	1.0	28.36	1.32	1.47
Rotational RMS residual / MHz	0.403	0.043	2.439	1.875
Iterations	5	3	25	8

```
parameter_class = CC_skeleton; mode=shared;
patterns = R(1,2)|R(1,6)|R(2,3)|R(2,45)|R(3,4)|R(3,10)|...

parameter_class = non_CC_stretches; mode=fixed;
patterns = all retained stretch combinations involving C-H, O-H, or C-O

parameter_class = Angles_fixed; mode=fixed; patterns=AAng
parameter_class = Torsions_fixed; mode=fixed; patterns=ATor
parameter_class = OOP_fixed; mode=fixed; patterns=AOop
```

For the two-class model, the script now gives primitive classes, not a molecule-specific list of GIC labels:

```
primitive_class = CC_skeleton;
patterns = R(1,2)|R(1,6)|R(2,3)|R(2,45)|...

primitive_class = CO_stretches;
patterns = R(7,13)|R(16,17)

primitive_class_min = 0.70
primitive_class_cross_max = 0.20
primitive_class_budget = auto
```

With the generated testosterone GIC basis, this gives 21 C–C-dominated GICs and two C–O-dominated GICs (`GIC020` and `GIC040`). The `auto` budget limits the number of independent class corrections to the number compatible with the available experimental observables; among competing declared classes, the most populated and highest-weighted classes are retained.

The predicate model freezes the complement of its compact C–C/C–O subset but keeps those directions independent. The C–C and C–O bond predicates are evaluated from the QC input geometry at run time and are therefore not hard-coded numerical constants in the script.

The important point is that the class is applied in the final working basis, not as a molecule-specific Z-matrix instruction. The input consists of Cartesian coordinates and the literature rotational data; the topology labels determine which retained GICs belong to the chemically declared classes.

The exact repository commands used for these diagnostics are archived in the testosterone example directory. They can be run from the repository root and read the QC Cartesian geometry, the published parent rotational constants, and the published vibrational corrections:

```
sh packages/matrix-morpheus/examples/semiexp/testosterone/
   run_testosterone_cc_class.sh
sh packages/matrix-morpheus/examples/semiexp/testosterone/
   run_testosterone_cc_co_classes.sh
sh packages/matrix-morpheus/examples/semiexp/testosterone/
   run_testosterone_predicates.sh
```

S12 Characteristics of the Test Cases

Table S28: Constraints and coordinate choices (symmetry-adapted internal or Cartesian coordinates, SAI and SAC, respectively) used for the benchmark calculations.

Case	Constraints	Coordinates
Glycolaldehyde	No fixed primitive constraints	SAI, ABC
Cyclopentadiene	No fixed primitive constraints	SAI and SAC, ABC
Nitrobenzene	No fixed primitive constraints	SAI, AB
p-EBN	Legacy frozen Z-matrix parameters	SAI, AB
Azulene	C–H bond lengths and selected angles	SAI, AB
Norcamphor	H-atom constraints and Kraitchman predicates	SAI and SAC, ABC
Camphor	BDPCS3 predicates; Kraitchman diagnostic only	SAI and SAC, ABC
Glycine I	Four generated QC-reference constraints	SAI, ABC
Glycine II	Five generated QC constraints	SAI, ABC
Maleic anhydride	No fixed primitive constraints	SAI and SAC, AB
Phthalic anhydride	Weighted r_e^{BO} predicates	SAI, AC
Succinic anhydride	Fixed local H-frame parameters	SAI, AC
Testosterone	C–C class; C–C/C–O classes; QC predicates	SAI, ABC